

Item Recommendation Reasoning Guide for the AI Team

Scope

- Source workbook: DataV3.xlsx .
- Requested item list: the screenshot prefixes were matched to the exact item codes below.
- Recommendation method: Active Demand with Active Variability .
- Forecast mode: Rules Only (no AutoGluon forecast blended in for this report).
- Seasonality: enabled, planning season Summer , so only April through September demand is considered relevant for the active seasonal window.
- Service levels used by the calculation: A=99% , B=92% , C=88% , D=84% , E=80% , X=80% .
- Location 1 is treated as the DC plus a stocking location. Locations 115 , 116 , 117 , 118 , 119 , 30 , and 40 are branches. Location 9999 is ignored.

Plain-English Rule Order

1. Ignore rows that should never drive replenishment: ignored statuses, Stock with no Max , and location 9999 .
2. Pick the history window. If year-over-year demand is between -33% and +33% , use both years. Otherwise use the last 12 months.
3. Apply seasonality. Because this report is for Summer , the usable months are only April through September inside the chosen history window.
4. Build the demand signal. In Active Demand with Active Variability , the mean and the standard deviation both ignore zero-demand months.

5. Blend in the product-group trend. The item mean is nudged up or down based on how the item's product group has been moving recently, but the adjustment is capped so it cannot overwhelm the item's own history.
6. Detect intermittent branch demand. For branch locations, if demand is too sparse or too concentrated in only one or two months, the model stops treating those few sales as steady branch demand. Instead it falls back to the inclusive seasonal mean and pushes the extra spike stock back to the DC.
7. Convert monthly demand into protection days. Protection days are `lead time + supplier order frequency`, with a floor of `28` days. If frequency is `Local`, the model treats it as `28` days.
8. Set the branch minimum. Branch mins are demand-only coverage. They do not hold large variability buffers.
9. Set the branch maximum. Branch maxes use the effective branch demand signal, but branch variability is not kept locally. That deviation is pooled into the DC.
10. Set the DC minimum. Location `1` only keeps enough min stock for its own demand between replenishment cycles.
11. Set the DC maximum. The DC/balancing location absorbs pooled branch deviation and intermittent spike stock, then is floored to the policy order-up-to level driven by lead time, frequency, and ABC service level.
12. Respect overrides. If `Override = Y`, the current max wins and the row is flagged as an override.

What the programmers should mirror

- Treat this as a hybrid of deterministic rules plus demand estimation, not as a pure black-box forecast.
- The most important branch-vs-DC behavior is: branches hold demand-only working stock, while the DC carries the pooled uncertainty.
- The most important outlier protection is the intermittent branch rule. That is what stops a branch from looking like it steadily needs `16` just because it sold `16` in one or two isolated months.
- If the AI team wants to learn from this engine, the best training targets are the final `recommended_min_amount` and `recommended_new_max`, while the best explanatory features are the intermediate fields shown in the item sections below.

Item-by-Item Reasoning

A-GAS REFRIGERANTS | R-134A-30LB-REFG | HFC134A

- Profile: ABC `A` at `99%` service level | season `Summer` | frequency `2 Weeks` (`14` days) | lead time `7` days | planning season `Summer`

- Reference window: 24 months | YOY -4.0% | rule: Year-over-year demand stayed within +/-33%, so both years are used.
- Product-group trend: Flat at 1.00x from Global product group
- Company total min: 27 -> 21 | company total max: 44 -> 115
- Pooled to DC: 53.50 | absorbed by DC: 53.50 | changed locations: 7 of 8

Location	Current Min	New Min	Current Max	New Max	Mean Used	Raw Mean	Total Dev	Pooled to DC	Absorbed by DC
1	3	9	12	77	9.55	9.55	13.09	0.00	53.50
115	2	1	3	4	4.27	4.27	3.12	7.00	0.00
116	7	2	9	9	8.67	8.67	6.13	14.00	0.00
117	4	2	5	8	8.33	8.33	4.99	11.00	0.00
118	4	2	5	6	5.67	5.67	2.46	6.00	0.00
119	3	1	4	1	0.50	2.00	0.50	3.50	0.00
30	1	2	2	5	4.91	4.91	3.34	8.00	0.00
40	3	2	4	5	4.40	4.40	2.25	4.00	0.00

Location reasoning:

- 1 : DC / balancing location. Current min/max 3 / 12 change to 9 / 77 . The DC min only covers location 1's own protected-cycle demand. The DC max absorbs 53.50 units of pooled branch uncertainty and is floored to policy order-up-to 39 . The protection window was floored up to 28 days from a raw 21 days.
- 115 : Branch location. Current min/max 2 / 3 change to 1 / 4 . The branch uses an effective demand basis of 4.27 and a variability amount of 3.12 to size the branch target, but the branch only keeps demand-only coverage. 7.00 units of deviation are intentionally not held here and are pooled to the DC instead. The protection window was floored up to 28 days from a raw 21 days.
- 116 : Branch location. Current min/max 7 / 9 change to 2 / 9 . The branch uses an effective demand basis of 8.67 and a variability amount of 6.13 to size the branch target, but the branch only keeps demand-only coverage. 14.00 units of deviation are intentionally not held here and are pooled to the DC instead. The protection window was floored up to 28 days from a raw 21 days.
- 117 : Branch location. Current min/max 4 / 5 change to 2 / 8 . The branch uses an effective demand basis of 8.33 and a variability amount of 4.99 to size the branch target, but the branch only keeps demand-only coverage. 11.00 units of deviation are intentionally not held here and are pooled to

the DC instead. The protection window was floored up to 28 days from a raw 21 days.

- 118 : Branch location. Current min/max 4 / 5 change to 2 / 6 . The branch uses an effective demand basis of 5.67 and a variability amount of 2.46 to size the branch target, but the branch only keeps demand-only coverage. 6.00 units of deviation are intentionally not held here and are pooled to the DC instead. The protection window was floored up to 28 days from a raw 21 days.

- 119 : Branch location. Current min/max 3 / 4 change to 1 / 1 . Intermittent branch protection fired, so the raw active-demand mean 2.00 was reduced to 0.50 because ADI is 4.0, the top two months account for 83% of scoped demand. 1.50 units of spike stock and 3.50 total branch deviation were pushed back to the DC. The protection window was floored up to 28 days from a raw 21 days.

- 30 : Branch location. Current min/max 1 / 2 change to 2 / 5 . The branch uses an effective demand basis of 4.91 and a variability amount of 3.34 to size the branch target, but the branch only keeps demand-only coverage. 8.00 units of deviation are intentionally not held here and are pooled to the DC instead. The protection window was floored up to 28 days from a raw 21 days.

- 40 : Branch location. Current min/max 3 / 4 change to 2 / 5 . The branch uses an effective demand basis of 4.40 and a variability amount of 2.25 to size the branch target, but the branch only keeps demand-only coverage. 4.00 units of deviation are intentionally not held here and are pooled to the DC instead. The protection window was floored up to 28 days from a raw 21 days.

A-GAS REFRIGERANTS | R-22-30LB-REFG | REFRIGERANT HCFC

- Profile: ABC A at 99% service level | season Summer | frequency 2 Weeks (14 days) | lead time 7 days | planning season Summer
- Reference window: 24 months | YOY -11.4% | rule: Year-over-year demand stayed within +/-33%, so both years are used.
- Product-group trend: Flat at 1.00x from No product-group trend
- Company total min: 16 -> 18 | company total max: 41 -> 90
- Pooled to DC: 44.00 | absorbed by DC: 44.00 | changed locations: 5 of 8

Location	Current Min	New Min	Current Max	New Max	Mean Used	Raw Mean	Total Dev	Pooled to DC	Absorbed by DC
1	6	7	24	58	7.50	7.50	6.11	0.00	44.00
115	1	1	2	2	2.00	2.00	0.89	2.00	0.00
116	2	3	3	9	8.70	8.70	4.44	10.00	0.00
117	1	2	2	6	5.50	5.50	3.23	8.00	0.00
118	2	1	3	3	2.73	2.73	1.68	4.00	0.00
119	1	1	2	2	1.57	1.57	0.71	2.00	0.00
30	1	1	2	4	3.40	3.40	2.86	5.00	0.00
40	2	2	3	6	6.18	6.18	5.71	13.00	0.00

Location reasoning:

- 1 : DC / balancing location. Current min/max 6 / 24 change to 7 / 58 . The DC min only covers location 1's own protected-cycle demand. The DC max absorbs 44.00 units of pooled branch uncertainty and is floored to policy order-up-to 22 . The protection window was floored up to 28 days from a raw 21 days.
- 115 : Branch location. Current min/max 1 / 2 change to 1 / 2 . The branch uses an effective demand basis of 2.00 and a variability amount of 0.89 to size the branch target, but the branch only keeps demand-only coverage. 2.00 units of deviation are intentionally not held here and are pooled to the DC instead. The protection window was floored up to 28 days from a raw 21 days.

- 116 : Branch location. Current min/max 2 / 3 change to 3 / 9 . The branch uses an effective demand basis of 8.70 and a variability amount of 4.44 to size the branch target, but the branch only keeps demand-only coverage. 10.00 units of deviation are intentionally not held here and are pooled to the DC instead. The protection window was floored up to 28 days from a raw 21 days.
- 117 : Branch location. Current min/max 1 / 2 change to 2 / 6 . The branch uses an effective demand basis of 5.50 and a variability amount of 3.23 to size the branch target, but the branch only keeps demand-only coverage. 8.00 units of deviation are intentionally not held here and are pooled to the DC instead. The protection window was floored up to 28 days from a raw 21 days.
- 118 : Branch location. Current min/max 2 / 3 change to 1 / 3 . The branch uses an effective demand basis of 2.73 and a variability amount of 1.68 to size the branch target, but the branch only keeps demand-only coverage. 4.00 units of deviation are intentionally not held here and are pooled to the DC instead. The protection window was floored up to 28 days from a raw 21 days.
- 119 : Branch location. Current min/max 1 / 2 change to 1 / 2 . The branch uses an effective demand basis of 1.57 and a variability amount of 0.71 to size the branch target, but the branch only keeps demand-only coverage. 2.00 units of deviation are intentionally not held here and are pooled to the DC instead. The protection window was floored up to 28 days from a raw 21 days.
- 30 : Branch location. Current min/max 1 / 2 change to 1 / 4 . The branch uses an effective demand basis of 3.40 and a variability amount of 2.86 to size the branch target, but the branch only keeps demand-only coverage. 5.00 units of deviation are intentionally not held here and are pooled to the DC instead. The protection window was floored up to 28 days from a raw 21 days.
- 40 : Branch location. Current min/max 2 / 3 change to 2 / 6 . The branch uses an effective demand basis of 6.18 and a variability amount of 5.71 to size the branch target, but the branch only keeps demand-only coverage. 13.00 units of deviation are intentionally not held here and are pooled to the DC instead. The protection window was floored up to 28 days from a raw 21 days.

ARKEMA, INC. | R-427A-25LB-REFG | R-22-REPLACEMENT

- Profile: ABC C at 88% service level | season Summer | frequency 2 Weeks (14 days) | lead time 14 days | planning season Summer
- Reference window: 12 months | YOY -46.3% | rule: Year-over-year demand moved outside +/-33%, so the model uses the last 12 months.
- Product-group trend: Flat at 0.99x from Global product group
- Company total min: 7 -> 4 | company total max: 33 -> 16
- Pooled to DC: 10.13 | absorbed by DC: 10.13 | changed locations: 3 of 8

Location	Current Min	New Min	Current Max	New Max	Mean Used	Raw Mean	Total Dev	Pooled to DC	Absorbed by DC
1	2	1	22	11	1.00	1.00	-0.01	0.00	10.13
115	0	1	1	1	0.17	1.00	0.17	0.83	0.00
116	1	0	2	1	0.00	0.00	0.00	0.00	0.00
117	0	0	0	0	0.00	0.00	0.00	0.00	0.00
118	4	1	6	1	0.83	2.50	1.95	4.65	0.00
119	0	0	1	1	0.00	0.00	0.00	0.00	0.00
30	0	1	1	1	0.83	2.50	1.95	4.65	0.00
40	0	0	0	0	0.00	0.00	0.00	0.00	0.00

Location reasoning:

- 1 : DC / balancing location. Current min/max 2 / 22 change to 1 / 11 . The DC min only covers location 1's own protected-cycle demand. The DC max absorbs 10.13 units of pooled branch uncertainty and is floored to policy order-up-to 1 .
- 115 : Branch location. Current min/max 0 / 1 change to 1 / 1 . Intermittent branch protection fired, so the raw active-demand mean 1.00 was reduced to 0.17 because it only sold in 1 of 6 scoped months, ADI is 6.0, the top two months account for 100% of scoped demand. 0.83 units of spike stock and 0.83 total branch deviation were pushed back to the DC.
- 116 : Branch location. Current min/max 1 / 2 change to 0 / 1 . The branch uses an effective demand basis of 0.00 and a variability amount of 0.00

to size the branch target, but the branch only keeps demand-only coverage. 0.00 units of deviation are intentionally not held here and are pooled to the DC instead.

- 117 : Branch location. Current min/max 0 / 0 change to 0 / 0 . The branch uses an effective demand basis of 0.00 and a variability amount of 0.00 to size the branch target, but the branch only keeps demand-only coverage. 0.00 units of deviation are intentionally not held here and are pooled to the DC instead.

- 118 : Branch location. Current min/max 4 / 6 change to 1 / 1 . Intermittent branch protection fired, so the raw active-demand mean 2.50 was reduced to 0.83 because it only sold in 2 of 6 scoped months, the top two months account for 100% of scoped demand. 1.65 units of spike stock and 4.65 total branch deviation were pushed back to the DC.

- 119 : Branch location. Current min/max 0 / 1 change to 0 / 1 . The branch uses an effective demand basis of 0.00 and a variability amount of 0.00 to size the branch target, but the branch only keeps demand-only coverage. 0.00 units of deviation are intentionally not held here and are pooled to the DC instead.

- 30 : Branch location. Current min/max 0 / 1 change to 1 / 1 . Intermittent branch protection fired, so the raw active-demand mean 2.50 was reduced to 0.83 because it only sold in 2 of 6 scoped months, the top two months account for 100% of scoped demand. 1.65 units of spike stock and 4.65 total branch deviation were pushed back to the DC.

- 40 : Branch location. Current min/max 0 / 0 change to 0 / 0 . The branch uses an effective demand basis of 0.00 and a variability amount of 0.00 to size the branch target, but the branch only keeps demand-only coverage. 0.00 units of deviation are intentionally not held here and are pooled to the DC instead.

CHEMOURS | R-417C-25LB-REFG | HOT SHOT TWO R-12 MP-39 R134A R500

- Profile: ABC C at 88% service level | season Summer | frequency 2 Weeks (14 days) | lead time 28 days | planning season Summer
- Reference window: 24 months | YOY -21.4% | rule: Year-over-year demand stayed within +/-33%, so both years are used.
- Product-group trend: Down at 0.85x from Supplier product group
- Company total min: 5 -> 7 | company total max: 28 -> 10
- Pooled to DC: 6.75 | absorbed by DC: 6.75 | changed locations: 4 of 8

Location	Current Min	New Min	Current Max	New Max	Mean Used	Raw Mean	Total Dev	Pooled to DC	Absorbed by DC
1	2	2	20	3	1.00	1.00	-0.15	0.00	6.75
115	0	1	1	1	0.25	1.50	0.92	2.06	0.00
116	1	1	2	1	0.25	3.00	0.21	2.34	0.00
117	0	0	0	0	0.00	0.00	0.00	0.00	0.00
118	1	1	2	2	1.25	1.25	0.56	1.00	0.00
119	0	0	1	1	0.00	0.00	0.00	0.00	0.00
30	1	1	2	1	0.25	1.00	0.21	0.64	0.00
40	0	1	0	1	0.17	1.00	0.14	0.71	0.00

Location reasoning:

- 1 : DC / balancing location. Current min/max 2 / 20 change to 2 / 3 . The DC min only covers location 1's own protected-cycle demand. The DC max absorbs 6.75 units of pooled branch uncertainty and is floored to policy order-up-to 2 .
- 115 : Branch location. Current min/max 0 / 1 change to 1 / 1 . Intermittent branch protection fired, so the raw active-demand mean 1.50 was reduced to 0.25 because it only sold in 2 of 12 scoped months, ADI is 6.0, the top two months account for 100% of scoped demand. 1.06 units of spike stock and 2.06 total branch deviation were pushed back to the DC.
- 116 : Branch location. Current min/max 1 / 2 change to 1 / 1 . Intermittent branch protection fired, so the raw active-demand mean 3.00 was

reduced to 0.25 because it only sold in 1 of 12 scoped months, ADI is 12.0, the top two months account for 100% of scoped demand. 2.34 units of spike stock and 2.34 total branch deviation were pushed back to the DC.

- 117 : Branch location. Current min/max 0 / 0 change to 0 / 0. The branch uses an effective demand basis of 0.00 and a variability amount of 0.00 to size the branch target, but the branch only keeps demand-only coverage. 0.00 units of deviation are intentionally not held here and are pooled to the DC instead.

- 118 : Branch location. Current min/max 1 / 2 change to 1 / 2. The branch uses an effective demand basis of 1.25 and a variability amount of 0.56 to size the branch target, but the branch only keeps demand-only coverage. 1.00 units of deviation are intentionally not held here and are pooled to the DC instead.

- 119 : Branch location. Current min/max 0 / 1 change to 0 / 1. The branch uses an effective demand basis of 0.00 and a variability amount of 0.00 to size the branch target, but the branch only keeps demand-only coverage. 0.00 units of deviation are intentionally not held here and are pooled to the DC instead.

- 30 : Branch location. Current min/max 1 / 2 change to 1 / 1. Intermittent branch protection fired, so the raw active-demand mean 1.00 was reduced to 0.25 because ADI is 4.0. 0.64 units of spike stock and 0.64 total branch deviation were pushed back to the DC.

- 40 : Branch location. Current min/max 0 / 0 change to 1 / 1. Intermittent branch protection fired, so the raw active-demand mean 1.00 was reduced to 0.17 because it only sold in 2 of 12 scoped months, ADI is 6.0, the top two months account for 100% of scoped demand. 0.71 units of spike stock and 0.71 total branch deviation were pushed back to the DC.

HUDSON TECHNOLOGIES COMPANY | R-407C-25LB-REFG | 407C SUVA REPLACE R-22

- Profile: ABC A at 99% service level | season Summer | frequency 2 Weeks (14 days) | lead time 28 days | planning season Summer
- Reference window: 24 months | YOY -16.7% | rule: Year-over-year demand stayed within +/-33%, so both years are used.
- Product-group trend: Down at 0.96x from Supplier product group
- Company total min: 34 -> 132 | company total max: 74 -> 282
- Pooled to DC: 139.00 | absorbed by DC: 139.00 | changed locations: 8 of 8

Location	Current Min	New Min	Current Max	New Max	Mean Used	Raw Mean	Total Dev	Pooled to DC	Absorbed by DC
1	9	44	38	151	33.00	33.00	18.00	0.00	139.00
115	4	9	5	14	9.92	9.92	5.27	15.00	0.00
116	4	11	6	16	11.50	11.50	4.97	13.00	0.00
117	5	20	7	29	21.75	21.75	8.47	24.00	0.00
118	4	12	6	18	13.25	13.25	6.47	18.00	0.00
119	1	7	2	10	7.42	7.42	6.47	18.00	0.00
30	4	14	6	21	15.33	15.33	9.35	26.00	0.00
40	3	15	4	23	16.83	16.83	9.54	25.00	0.00

Location reasoning:

- 1 : DC / balancing location. Current min/max 9 / 38 change to 44 / 151 . The DC min only covers location 1's own protected-cycle demand. The DC max absorbs 139.00 units of pooled branch uncertainty and is floored to policy order-up-to 94 .
- 115 : Branch location. Current min/max 4 / 5 change to 9 / 14 . The branch uses an effective demand basis of 9.92 and a variability amount of 5.27 to size the branch target, but the branch only keeps demand-only coverage. 15.00 units of deviation are intentionally not held here and are pooled to the DC instead.
- 116 : Branch location. Current min/max 4 / 6 change to 11 / 16 . The branch uses an effective demand basis of 11.50 and a variability amount of

- 4.97 to size the branch target, but the branch only keeps demand-only coverage. 13.00 units of deviation are intentionally not held here and are pooled to the DC instead.
- 117 : Branch location. Current min/max 5 / 7 change to 20 / 29 . The branch uses an effective demand basis of 21.75 and a variability amount of 8.47 to size the branch target, but the branch only keeps demand-only coverage. 24.00 units of deviation are intentionally not held here and are pooled to the DC instead.
 - 118 : Branch location. Current min/max 4 / 6 change to 12 / 18 . The branch uses an effective demand basis of 13.25 and a variability amount of 6.47 to size the branch target, but the branch only keeps demand-only coverage. 18.00 units of deviation are intentionally not held here and are pooled to the DC instead.
 - 119 : Branch location. Current min/max 1 / 2 change to 7 / 10 . The branch uses an effective demand basis of 7.42 and a variability amount of 6.47 to size the branch target, but the branch only keeps demand-only coverage. 18.00 units of deviation are intentionally not held here and are pooled to the DC instead.
 - 30 : Branch location. Current min/max 4 / 6 change to 14 / 21 . The branch uses an effective demand basis of 15.33 and a variability amount of 9.35 to size the branch target, but the branch only keeps demand-only coverage. 26.00 units of deviation are intentionally not held here and are pooled to the DC instead.
 - 40 : Branch location. Current min/max 3 / 4 change to 15 / 23 . The branch uses an effective demand basis of 16.83 and a variability amount of 9.54 to size the branch target, but the branch only keeps demand-only coverage. 25.00 units of deviation are intentionally not held here and are pooled to the DC instead.

HUDSON TECHNOLOGIES COMPANY | R-410A-25LB-REFG | REFRIGERANT HFC

- Profile: ABC A at 99% service level | season Summer | frequency 2 Weeks (14 days) | lead time 28 days | planning season Summer
- Reference window: 24 months | YOY -25.0% | rule: Year-over-year demand stayed within +/-33%, so both years are used.
- Product-group trend: Flat at 1.00x from No product-group trend
- Company total min: 192 -> 596 | company total max: 356 -> 1,407
- Pooled to DC: 726.00 | absorbed by DC: 726.00 | changed locations: 8 of 8

Location	Current Min	New Min	Current Max	New Max	Mean Used	Raw Mean	Total Dev	Pooled to DC	Absorbed by DC
1	36	126	159	702	90.83	90.83	38.18	0.00	726.00
115	19	70	24	104	75.25	75.25	45.34	124.00	0.00
116	43	98	54	147	105.83	105.83	80.29	219.00	0.00
117	25	65	32	98	70.58	70.58	14.38	40.00	0.00
118	10	33	13	50	35.58	35.58	32.66	90.00	0.00
119	16	29	20	43	31.08	31.08	23.98	66.00	0.00
30	20	91	25	137	98.75	98.75	34.13	94.00	0.00
40	23	84	29	126	90.83	90.83	34.08	93.00	0.00

Location reasoning:

- 1 : DC / balancing location. Current min/max 36 / 159 change to 126 / 702 . The DC min only covers location 1's own protected-cycle demand. The DC max absorbs 726.00 units of pooled branch uncertainty and is floored to policy order-up-to 231 .
- 115 : Branch location. Current min/max 19 / 24 change to 70 / 104 . The branch uses an effective demand basis of 75.25 and a variability amount of 45.34 to size the branch target, but the branch only keeps demand-only coverage. 124.00 units of deviation are intentionally not held here and are pooled to the DC instead.
- 116 : Branch location. Current min/max 43 / 54 change to 98 / 147 . The branch uses an effective demand basis of 105.83 and a variability amount

of 80.29 to size the branch target, but the branch only keeps demand-only coverage. 219.00 units of deviation are intentionally not held here and are pooled to the DC instead.

- 117 : Branch location. Current min/max 25 / 32 change to 65 / 98 . The branch uses an effective demand basis of 70.58 and a variability amount of 14.38 to size the branch target, but the branch only keeps demand-only coverage. 40.00 units of deviation are intentionally not held here and are pooled to the DC instead.

- 118 : Branch location. Current min/max 10 / 13 change to 33 / 50 . The branch uses an effective demand basis of 35.58 and a variability amount of 32.66 to size the branch target, but the branch only keeps demand-only coverage. 90.00 units of deviation are intentionally not held here and are pooled to the DC instead.

- 119 : Branch location. Current min/max 16 / 20 change to 29 / 43 . The branch uses an effective demand basis of 31.08 and a variability amount of 23.98 to size the branch target, but the branch only keeps demand-only coverage. 66.00 units of deviation are intentionally not held here and are pooled to the DC instead.

- 30 : Branch location. Current min/max 20 / 25 change to 91 / 137 . The branch uses an effective demand basis of 98.75 and a variability amount of 34.13 to size the branch target, but the branch only keeps demand-only coverage. 94.00 units of deviation are intentionally not held here and are pooled to the DC instead.

- 40 : Branch location. Current min/max 23 / 29 change to 84 / 126 . The branch uses an effective demand basis of 90.83 and a variability amount of 34.08 to size the branch target, but the branch only keeps demand-only coverage. 93.00 units of deviation are intentionally not held here and are pooled to the DC instead.

HUDSON TECHNOLOGIES COMPANY | R-422B-25LB-REFG | NU-22B REFRIGERANT R-22 REPLMNT

- Profile: ABC B at 92% service level | season Summer | frequency 2 Weeks (14 days) | lead time 28 days | planning season Summer
- Reference window: 24 months | YOY -5.6% | rule: Year-over-year demand stayed within +/-33%, so both years are used.
- Product-group trend: Down at 0.96x from Supplier product group
- Company total min: 7 -> 21 | company total max: 20 -> 30
- Pooled to DC: 11.96 | absorbed by DC: 11.96 | changed locations: 6 of 8

Location	Current Min	New Min	Current Max	New Max	Mean Used	Raw Mean	Total Dev	Pooled to DC	Absorbed by DC
1	2	5	8	10	3.64	3.64	2.95	0.00	11.96
115	1	2	2	2	1.20	1.20	0.60	1.00	0.00
116	0	2	1	3	1.80	1.80	0.57	1.00	0.00
117	0	1	1	1	0.33	1.33	0.90	1.96	0.00
118	0	2	1	2	1.33	1.33	0.80	1.00	0.00
119	2	3	3	4	2.33	2.33	1.75	2.00	0.00
30	1	2	2	3	2.00	2.00	0.59	1.00	0.00
40	1	4	2	5	3.67	3.67	1.94	4.00	0.00

Location reasoning:

- 1 : DC / balancing location. Current min/max 2 / 8 change to 5 / 10 . The DC min only covers location 1's own protected-cycle demand. The DC max absorbs 11.96 units of pooled branch uncertainty and is floored to policy order-up-to 9 .
- 115 : Branch location. Current min/max 1 / 2 change to 2 / 2 . The branch uses an effective demand basis of 1.20 and a variability amount of 0.60 to size the branch target, but the branch only keeps demand-only coverage. 1.00 units of deviation are intentionally not held here and are pooled to the DC instead.
- 116 : Branch location. Current min/max 0 / 1 change to 2 / 3 . The branch uses an effective demand basis of 1.80 and a variability amount of 0.57

to size the branch target, but the branch only keeps demand-only coverage. 1.00 units of deviation are intentionally not held here and are pooled to the DC instead.

- 117 : Branch location. Current min/max 0 / 1 change to 1 / 1. Intermittent branch protection fired, so the raw active-demand mean 1.33 was reduced to 0.33 because ADI is 4.0, the top two months account for 75% of scoped demand. 0.96 units of spike stock and 1.96 total branch deviation were pushed back to the DC.

- 118 : Branch location. Current min/max 0 / 1 change to 2 / 2. The branch uses an effective demand basis of 1.33 and a variability amount of 0.80 to size the branch target, but the branch only keeps demand-only coverage. 1.00 units of deviation are intentionally not held here and are pooled to the DC instead.

- 119 : Branch location. Current min/max 2 / 3 change to 3 / 4. The branch uses an effective demand basis of 2.33 and a variability amount of 1.75 to size the branch target, but the branch only keeps demand-only coverage. 2.00 units of deviation are intentionally not held here and are pooled to the DC instead.

- 30 : Branch location. Current min/max 1 / 2 change to 2 / 3. The branch uses an effective demand basis of 2.00 and a variability amount of 0.59 to size the branch target, but the branch only keeps demand-only coverage. 1.00 units of deviation are intentionally not held here and are pooled to the DC instead.

- 40 : Branch location. Current min/max 1 / 2 change to 4 / 5. The branch uses an effective demand basis of 3.67 and a variability amount of 1.94 to size the branch target, but the branch only keeps demand-only coverage. 4.00 units of deviation are intentionally not held here and are pooled to the DC instead.

HUDSON TECHNOLOGIES COMPANY | R-422D-25LB-REFG | MO29 REFRIGERANT R-22 REPLMNT

- Profile: ABC B at 92% service level | season Summer | frequency 2 Weeks (14 days) | lead time 28 days | planning season Summer
- Reference window: 24 months | YOY 12.1% | rule: Year-over-year demand stayed within +/-33%, so both years are used.
- Product-group trend: Down at 0.96x from Supplier product group
- Company total min: 6 -> 18 | company total max: 21 -> 35
- Pooled to DC: 13.96 | absorbed by DC: 13.96 | changed locations: 4 of 8

Location	Current Min	New Min	Current Max	New Max	Mean Used	Raw Mean	Total Dev	Pooled to DC	Absorbed by DC
1	1	8	11	23	6.00	6.00	8.42	0.00	13.96
115	0	0	0	0	0.00	0.00	0.00	0.00	0.00
116	0	1	1	1	0.25	1.00	0.24	0.72	0.00
117	3	2	4	2	1.17	3.50	3.23	7.24	0.00
118	1	2	2	2	1.20	1.20	0.60	1.00	0.00
119	0	1	1	2	1.00	1.00	-0.04	0.00	0.00
30	0	0	0	0	0.00	0.00	0.00	0.00	0.00
40	1	4	2	5	3.60	3.60	3.05	5.00	0.00

Location reasoning:

- 1 : DC / balancing location. Current min/max 1 / 11 change to 8 / 23 . The DC min only covers location 1's own protected-cycle demand. The DC max absorbs 13.96 units of pooled branch uncertainty and is floored to policy order-up-to 23 .
- 115 : Branch location. Current min/max 0 / 0 change to 0 / 0 . The branch uses an effective demand basis of 0.00 and a variability amount of 0.00 to size the branch target, but the branch only keeps demand-only coverage. 0.00 units of deviation are intentionally not held here and are pooled to the DC instead.
- 116 : Branch location. Current min/max 0 / 1 change to 1 / 1 . Intermittent branch protection fired, so the raw active-demand mean 1.00 was

reduced to 0.25 because ADI is 4.0. 0.72 units of spike stock and 0.72 total branch deviation were pushed back to the DC.

- 117 : Branch location. Current min/max 3 / 4 change to 2 / 2. Intermittent branch protection fired, so the raw active-demand mean 3.50 was reduced to 1.17 because the top two months account for 79% of scoped demand. 2.24 units of spike stock and 7.24 total branch deviation were pushed back to the DC.

- 118 : Branch location. Current min/max 1 / 2 change to 2 / 2. The branch uses an effective demand basis of 1.20 and a variability amount of 0.60 to size the branch target, but the branch only keeps demand-only coverage. 1.00 units of deviation are intentionally not held here and are pooled to the DC instead.

- 119 : Branch location. Current min/max 0 / 1 change to 1 / 2. The branch uses an effective demand basis of 1.00 and a variability amount of -0.04 to size the branch target, but the branch only keeps demand-only coverage. 0.00 units of deviation are intentionally not held here and are pooled to the DC instead.

- 30 : Branch location. Current min/max 0 / 0 change to 0 / 0. The branch uses an effective demand basis of 0.00 and a variability amount of 0.00 to size the branch target, but the branch only keeps demand-only coverage. 0.00 units of deviation are intentionally not held here and are pooled to the DC instead.

- 40 : Branch location. Current min/max 1 / 2 change to 4 / 5. The branch uses an effective demand basis of 3.60 and a variability amount of 3.05 to size the branch target, but the branch only keeps demand-only coverage. 5.00 units of deviation are intentionally not held here and are pooled to the DC instead.

HUDSON TECHNOLOGIES COMPANY | R-448A-25LB-REFG | REFRIGERANT HFO 25LB CYLINDER

- Profile: ABC A at 99% service level | season Summer | frequency 2 Weeks (14 days) | lead time 28 days | planning season Summer
- Reference window: 24 months | YOY -9.8% | rule: Year-over-year demand stayed within +/-33%, so both years are used.
- Product-group trend: Down at 0.96x from Supplier product group
- Company total min: 23 -> 29 | company total max: 44 -> 63
- Pooled to DC: 33.88 | absorbed by DC: 33.88 | changed locations: 6 of 8

Location	Current Min	New Min	Current Max	New Max	Mean Used	Raw Mean	Total Dev	Pooled to DC	Absorbed by DC
1	4	6	15	32	4.40	4.40	3.29	0.00	33.88
115	1	2	2	2	1.29	1.29	0.72	2.00	0.00
116	1	2	2	3	2.11	2.11	1.56	4.00	0.00
117	5	5	7	7	4.75	4.75	3.21	10.00	0.00
118	9	8	12	11	8.25	8.25	3.06	9.00	0.00
119	1	1	2	1	0.08	1.00	0.08	0.88	0.00
30	1	2	2	3	2.00	2.00	1.15	3.00	0.00
40	1	3	2	4	2.75	2.75	1.41	5.00	0.00

Location reasoning:

- 1 : DC / balancing location. Current min/max 4 / 15 change to 6 / 32 . The DC min only covers location 1's own protected-cycle demand. The DC max absorbs 33.88 units of pooled branch uncertainty and is floored to policy order-up-to 15 .
- 115 : Branch location. Current min/max 1 / 2 change to 2 / 2 . The branch uses an effective demand basis of 1.29 and a variability amount of 0.72 to size the branch target, but the branch only keeps demand-only coverage. 2.00 units of deviation are intentionally not held here and are pooled to the DC instead.
- 116 : Branch location. Current min/max 1 / 2 change to 2 / 3 . The branch uses an effective demand basis of 2.11 and a variability amount of 1.56

to size the branch target, but the branch only keeps demand-only coverage. 4.00 units of deviation are intentionally not held here and are pooled to the DC instead.

- 117 : Branch location. Current min/max 5 / 7 change to 5 / 7 . The branch uses an effective demand basis of 4.75 and a variability amount of 3.21 to size the branch target, but the branch only keeps demand-only coverage. 10.00 units of deviation are intentionally not held here and are pooled to the DC instead.

- 118 : Branch location. Current min/max 9 / 12 change to 8 / 11 . The branch uses an effective demand basis of 8.25 and a variability amount of 3.06 to size the branch target, but the branch only keeps demand-only coverage. 9.00 units of deviation are intentionally not held here and are pooled to the DC instead.

- 119 : Branch location. Current min/max 1 / 2 change to 1 / 1 . Intermittent branch protection fired, so the raw active-demand mean 1.00 was reduced to 0.08 because it only sold in 1 of 12 scoped months, ADI is 12.0, the top two months account for 100% of scoped demand. 0.88 units of spike stock and 0.88 total branch deviation were pushed back to the DC.

- 30 : Branch location. Current min/max 1 / 2 change to 2 / 3 . The branch uses an effective demand basis of 2.00 and a variability amount of 1.15 to size the branch target, but the branch only keeps demand-only coverage. 3.00 units of deviation are intentionally not held here and are pooled to the DC instead.

- 40 : Branch location. Current min/max 1 / 2 change to 3 / 4 . The branch uses an effective demand basis of 2.75 and a variability amount of 1.41 to size the branch target, but the branch only keeps demand-only coverage. 5.00 units of deviation are intentionally not held here and are pooled to the DC instead.

HUDSON TECHNOLOGIES COMPANY | R-449A-25LB-REFG | OPTEON XP40

- Profile: ABC B at 92% service level | season Summer | frequency 2 Weeks (14 days) | lead time 28 days | planning season Summer
- Reference window: 24 months | YOY -30.0% | rule: Year-over-year demand stayed within +/-33%, so both years are used.
- Product-group trend: Down at 0.96x from Supplier product group
- Company total min: 11 -> 15 | company total max: 40 -> 24
- Pooled to DC: 11.28 | absorbed by DC: 11.28 | changed locations: 6 of 8

Location	Current Min	New Min	Current Max	New Max	Mean Used	Raw Mean	Total Dev	Pooled to DC	Absorbed by DC
1	3	5	25	11	3.50	3.50	2.26	0.00	11.28
115	2	1	3	1	0.25	1.00	0.24	0.72	0.00
116	1	2	2	3	1.83	1.83	0.74	2.00	0.00
117	1	1	2	1	0.17	1.00	0.16	0.80	0.00
118	2	1	3	2	1.00	1.00	1.37	2.00	0.00
119	0	1	1	1	0.17	2.00	0.16	1.76	0.00
30	1	2	2	3	2.20	2.20	1.15	2.00	0.00
40	1	2	2	2	1.50	1.50	1.20	2.00	0.00

Location reasoning:

- 1 : DC / balancing location. Current min/max 3 / 25 change to 5 / 11 . The DC min only covers location 1's own protected-cycle demand. The DC max absorbs 11.28 units of pooled branch uncertainty and is floored to policy order-up-to 8 .
- 115 : Branch location. Current min/max 2 / 3 change to 1 / 1 . Intermittent branch protection fired, so the raw active-demand mean 1.00 was reduced to 0.25 because ADI is 4.0. 0.72 units of spike stock and 0.72 total branch deviation were pushed back to the DC.
- 116 : Branch location. Current min/max 1 / 2 change to 2 / 3 . The branch uses an effective demand basis of 1.83 and a variability amount of 0.74 to size the branch target, but the branch only keeps demand-only coverage. 2.00 units of deviation are intentionally not held here and are pooled to the

DC instead.

- 117 : Branch location. Current min/max 1 / 2 change to 1 / 1 . Intermittent branch protection fired, so the raw active-demand mean 1.00 was reduced to 0.17 because it only sold in 2 of 12 scoped months, ADI is 6.0, the top two months account for 100% of scoped demand. 0.80 units of spike stock and 0.80 total branch deviation were pushed back to the DC.
- 118 : Branch location. Current min/max 2 / 3 change to 1 / 2 . The branch uses an effective demand basis of 1.00 and a variability amount of 1.37 to size the branch target, but the branch only keeps demand-only coverage. 2.00 units of deviation are intentionally not held here and are pooled to the DC instead.
- 119 : Branch location. Current min/max 0 / 1 change to 1 / 1 . Intermittent branch protection fired, so the raw active-demand mean 2.00 was reduced to 0.17 because it only sold in 1 of 12 scoped months, ADI is 12.0, the top two months account for 100% of scoped demand. 1.76 units of spike stock and 1.76 total branch deviation were pushed back to the DC.
- 30 : Branch location. Current min/max 1 / 2 change to 2 / 3 . The branch uses an effective demand basis of 2.20 and a variability amount of 1.15 to size the branch target, but the branch only keeps demand-only coverage. 2.00 units of deviation are intentionally not held here and are pooled to the DC instead.
- 40 : Branch location. Current min/max 1 / 2 change to 2 / 2 . The branch uses an effective demand basis of 1.50 and a variability amount of 1.20 to size the branch target, but the branch only keeps demand-only coverage. 2.00 units of deviation are intentionally not held here and are pooled to the DC instead.

NU-CALGON | R-290-10.6OZ-REFG | R-290-10.6OZ CAN

- Profile: ABC B at 92% service level | season Summer | frequency 2 Weeks (14 days) | lead time 14 days | planning season Summer
- Reference window: 24 months | YOY 14.9% | rule: Year-over-year demand stayed within +/-33%, so both years are used.
- Product-group trend: Up at 1.11x from Supplier product group
- Company total min: 43 -> 21 | company total max: 56 -> 56
- Pooled to DC: 20.01 | absorbed by DC: 20.01 | changed locations: 6 of 8

Location	Current Min	New Min	Current Max	New Max	Mean Used	Raw Mean	Total Dev	Pooled to DC	Absorbed by DC
1	11	5	14	28	4.73	4.73	3.30	0.00	20.01
115	7	3	8	5	4.73	4.73	3.16	4.00	0.00
116	8	3	10	5	4.58	4.58	2.22	3.00	0.00
117	2	2	3	4	3.11	3.11	2.81	3.00	0.00
118	7	3	9	6	5.36	5.36	2.22	3.00	0.00
119	0	1	1	1	0.08	1.00	0.09	1.01	0.00
30	5	2	7	3	2.64	2.64	1.35	2.00	0.00
40	3	2	4	4	3.73	3.73	2.95	4.00	0.00

Location reasoning:

- 1 : DC / balancing location. Current min/max 11 / 14 change to 5 / 28 . The DC min only covers location 1's own protected-cycle demand. The DC max absorbs 20.01 units of pooled branch uncertainty and is floored to policy order-up-to 9 .
- 115 : Branch location. Current min/max 7 / 8 change to 3 / 5 . The branch uses an effective demand basis of 4.73 and a variability amount of 3.16 to size the branch target, but the branch only keeps demand-only coverage. 4.00 units of deviation are intentionally not held here and are pooled to the DC instead.
- 116 : Branch location. Current min/max 8 / 10 change to 3 / 5 . The branch uses an effective demand basis of 4.58 and a variability amount of

2.22 to size the branch target, but the branch only keeps demand-only coverage. 3.00 units of deviation are intentionally not held here and are pooled to the DC instead.

- 117 : Branch location. Current min/max 2 / 3 change to 2 / 4 . The branch uses an effective demand basis of 3.11 and a variability amount of 2.81 to size the branch target, but the branch only keeps demand-only coverage. 3.00 units of deviation are intentionally not held here and are pooled to the DC instead.

- 118 : Branch location. Current min/max 7 / 9 change to 3 / 6 . The branch uses an effective demand basis of 5.36 and a variability amount of 2.22 to size the branch target, but the branch only keeps demand-only coverage. 3.00 units of deviation are intentionally not held here and are pooled to the DC instead.

- 119 : Branch location. Current min/max 0 / 1 change to 1 / 1 . Intermittent branch protection fired, so the raw active-demand mean 1.00 was reduced to 0.08 because it only sold in 1 of 12 scoped months, ADI is 12.0, the top two months account for 100% of scoped demand. 1.01 units of spike stock and 1.01 total branch deviation were pushed back to the DC.

- 30 : Branch location. Current min/max 5 / 7 change to 2 / 3 . The branch uses an effective demand basis of 2.64 and a variability amount of 1.35 to size the branch target, but the branch only keeps demand-only coverage. 2.00 units of deviation are intentionally not held here and are pooled to the DC instead.

- 40 : Branch location. Current min/max 3 / 4 change to 2 / 4 . The branch uses an effective demand basis of 3.73 and a variability amount of 2.95 to size the branch target, but the branch only keeps demand-only coverage. 4.00 units of deviation are intentionally not held here and are pooled to the DC instead.

NU-CALGON | R-600-10.6OZ-REFG | R-600 10.6OZ CAN

- Profile: ABC C at 88% service level | season Summer | frequency 2 Weeks (14 days) | lead time 14 days | planning season Summer
- Reference window: 24 months | YOY 26.9% | rule: Year-over-year demand stayed within +/-33%, so both years are used.
- Product-group trend: Up at 1.11x from Supplier product group
- Company total min: 13 -> 10 | company total max: 20 -> 26
- Pooled to DC: 10.22 | absorbed by DC: 10.22 | changed locations: 2 of 8

Location	Current Min	New Min	Current Max	New Max	Mean Used	Raw Mean	Total Dev	Pooled to DC	Absorbed by DC
1	3	3	4	14	2.89	2.89	1.89	0.00	10.22
115	0	0	0	0	0.00	0.00	0.00	0.00	0.00
116	5	2	7	3	2.71	2.71	1.11	2.00	0.00
117	3	2	4	4	3.62	3.62	2.45	3.00	0.00
118	1	1	2	2	1.08	3.25	2.26	4.40	0.00
119	0	1	1	1	0.25	1.00	0.28	0.83	0.00
30	0	0	0	0	0.00	0.00	0.00	0.00	0.00
40	1	1	2	2	1.25	1.25	0.88	0.00	0.00

Location reasoning:

- 1 : DC / balancing location. Current min/max 3 / 4 change to 3 / 14 . The DC min only covers location 1's own protected-cycle demand. The DC max absorbs 10.22 units of pooled branch uncertainty and is floored to policy order-up-to 5 .
- 115 : Branch location. Current min/max 0 / 0 change to 0 / 0 . The branch uses an effective demand basis of 0.00 and a variability amount of 0.00 to size the branch target, but the branch only keeps demand-only coverage. 0.00 units of deviation are intentionally not held here and are pooled to the DC instead.
- 116 : Branch location. Current min/max 5 / 7 change to 2 / 3 . The branch uses an effective demand basis of 2.71 and a variability amount of 1.11

to size the branch target, but the branch only keeps demand-only coverage. 2.00 units of deviation are intentionally not held here and are pooled to the DC instead.

- 117 : Branch location. Current min/max 3 / 4 change to 2 / 4. The branch uses an effective demand basis of 3.62 and a variability amount of 2.45 to size the branch target, but the branch only keeps demand-only coverage. 3.00 units of deviation are intentionally not held here and are pooled to the DC instead.

- 118 : Branch location. Current min/max 1 / 2 change to 1 / 2. Intermittent branch protection fired, so the raw active-demand mean 3.25 was reduced to 1.08 because the top two months account for 77% of scoped demand. 2.40 units of spike stock and 4.40 total branch deviation were pushed back to the DC.

- 119 : Branch location. Current min/max 0 / 1 change to 1 / 1. Intermittent branch protection fired, so the raw active-demand mean 1.00 was reduced to 0.25 because ADI is 4.0. 0.83 units of spike stock and 0.83 total branch deviation were pushed back to the DC.

- 30 : Branch location. Current min/max 0 / 0 change to 0 / 0. The branch uses an effective demand basis of 0.00 and a variability amount of 0.00 to size the branch target, but the branch only keeps demand-only coverage. 0.00 units of deviation are intentionally not held here and are pooled to the DC instead.

- 40 : Branch location. Current min/max 1 / 2 change to 1 / 2. The branch uses an effective demand basis of 1.25 and a variability amount of 0.88 to size the branch target, but the branch only keeps demand-only coverage. 0.00 units of deviation are intentionally not held here and are pooled to the DC instead.

RGAS LLC | R-407A-25LB-REFG | REFG R404A/R507 REPLACEMENT

- Profile: ABC C at 88% service level | season Summer | frequency 2 Weeks (14 days) | lead time 7 days | planning season Summer
- Reference window: 24 months | YOY 6.0% | rule: Year-over-year demand stayed within +/-33%, so both years are used.
- Product-group trend: Flat at 1.00x from Global product group
- Company total min: 6 -> 8 | company total max: 21 -> 21
- Pooled to DC: 11.27 | absorbed by DC: 11.27 | changed locations: 2 of 8

Location	Current Min	New Min	Current Max	New Max	Mean Used	Raw Mean	Total Dev	Pooled to DC	Absorbed by DC
1	2	1	10	11	1.00	1.00	0.00	0.00	11.27
115	1	1	2	2	1.33	1.33	0.85	0.00	0.00
116	0	1	1	1	0.08	1.00	0.08	0.92	0.00
117	1	1	2	1	0.08	1.00	0.08	0.92	0.00
118	2	1	3	3	3.10	3.10	2.62	3.00	0.00
119	0	1	1	1	0.25	1.50	0.96	2.25	0.00
30	0	1	1	1	0.17	2.00	0.17	1.84	0.00
40	0	1	1	1	0.67	2.00	0.82	2.34	0.00

Location reasoning:

- 1 : DC / balancing location. Current min/max 2 / 10 change to 1 / 11 . The DC min only covers location 1's own protected-cycle demand. The DC max absorbs 11.27 units of pooled branch uncertainty and is floored to policy order-up-to 1 . The protection window was floored up to 28 days from a raw 21 days.
- 115 : Branch location. Current min/max 1 / 2 change to 1 / 2 . The branch uses an effective demand basis of 1.33 and a variability amount of 0.85 to size the branch target, but the branch only keeps demand-only coverage. 0.00 units of deviation are intentionally not held here and are pooled to the DC instead. The protection window was floored up to 28 days from a raw 21 days.

- 116 : Branch location. Current min/max 0 / 1 change to 1 / 1 . Intermittent branch protection fired, so the raw active-demand mean 1.00 was reduced to 0.08 because it only sold in 1 of 12 scoped months, ADI is 12.0, the top two months account for 100% of scoped demand. 0.92 units of spike stock and 0.92 total branch deviation were pushed back to the DC. The protection window was floored up to 28 days from a raw 21 days.
- 117 : Branch location. Current min/max 1 / 2 change to 1 / 1 . Intermittent branch protection fired, so the raw active-demand mean 1.00 was reduced to 0.08 because it only sold in 1 of 12 scoped months, ADI is 12.0, the top two months account for 100% of scoped demand. 0.92 units of spike stock and 0.92 total branch deviation were pushed back to the DC. The protection window was floored up to 28 days from a raw 21 days.
- 118 : Branch location. Current min/max 2 / 3 change to 1 / 3 . The branch uses an effective demand basis of 3.10 and a variability amount of 2.62 to size the branch target, but the branch only keeps demand-only coverage. 3.00 units of deviation are intentionally not held here and are pooled to the DC instead. The protection window was floored up to 28 days from a raw 21 days.
- 119 : Branch location. Current min/max 0 / 1 change to 1 / 1 . Intermittent branch protection fired, so the raw active-demand mean 1.50 was reduced to 0.25 because it only sold in 2 of 12 scoped months, ADI is 6.0, the top two months account for 100% of scoped demand. 1.25 units of spike stock and 2.25 total branch deviation were pushed back to the DC. The protection window was floored up to 28 days from a raw 21 days.
- 30 : Branch location. Current min/max 0 / 1 change to 1 / 1 . Intermittent branch protection fired, so the raw active-demand mean 2.00 was reduced to 0.17 because it only sold in 1 of 12 scoped months, ADI is 12.0, the top two months account for 100% of scoped demand. 1.84 units of spike stock and 1.84 total branch deviation were pushed back to the DC. The protection window was floored up to 28 days from a raw 21 days.
- 40 : Branch location. Current min/max 0 / 1 change to 1 / 1 . Intermittent branch protection fired, so the raw active-demand mean 2.00 was reduced to 0.67 because the top two months account for 75% of scoped demand. 1.34 units of spike stock and 2.34 total branch deviation were pushed back to the DC. The protection window was floored up to 28 days from a raw 21 days.

iGAS USA, inc | R-32-20LB-REFG | REFRIGERANT R32

- Profile: ABC A at 99% service level | season Summer | frequency 2 Weeks (14 days) | lead time 14 days | planning season Summer
- Reference window: 12 months | YOY 311.1% | rule: Year-over-year demand moved outside +/-33%, so the model uses the last 12 months.
- Product-group trend: Flat at 1.01x from Supplier product group
- Company total min: 17 -> 14 | company total max: 61 -> 59
- Pooled to DC: 33.77 | absorbed by DC: 33.77 | changed locations: 3 of 8

Location	Current Min	New Min	Current Max	New Max	Mean Used	Raw Mean	Total Dev	Pooled to DC	Absorbed by DC
1	3	3	40	39	3.20	3.20	2.50	0.00	33.77
115	1	1	2	2	1.17	1.75	1.13	2.59	0.00
116	3	2	4	4	3.67	3.67	1.76	5.00	0.00
117	1	1	2	2	1.80	1.80	1.12	3.00	0.00
118	2	1	3	1	1.00	2.00	1.74	5.01	0.00
119	1	1	2	2	1.17	2.33	1.33	3.17	0.00
30	3	3	4	5	4.80	4.80	3.60	8.00	0.00
40	3	2	4	4	4.17	4.17	3.32	7.00	0.00

Location reasoning:

- 1 : DC / balancing location. Current min/max 3 / 40 change to 3 / 39 . The DC min only covers location 1's own protected-cycle demand. The DC max absorbs 33.77 units of pooled branch uncertainty and is floored to policy order-up-to 9 .
- 115 : Branch location. Current min/max 1 / 2 change to 1 / 2 . Intermittent branch protection fired, so the raw active-demand mean 1.75 was reduced to 1.17 because the top two months account for 71% of scoped demand. 0.59 units of spike stock and 2.59 total branch deviation were pushed back to the DC.
- 116 : Branch location. Current min/max 3 / 4 change to 2 / 4 . The branch uses an effective demand basis of 3.67 and a variability amount of 1.76

to size the branch target, but the branch only keeps demand-only coverage. 5.00 units of deviation are intentionally not held here and are pooled to the DC instead.

- 117 : Branch location. Current min/max 1 / 2 change to 1 / 2 . The branch uses an effective demand basis of 1.80 and a variability amount of 1.12 to size the branch target, but the branch only keeps demand-only coverage. 3.00 units of deviation are intentionally not held here and are pooled to the DC instead.

- 118 : Branch location. Current min/max 2 / 3 change to 1 / 1 . Intermittent branch protection fired, so the raw active-demand mean 2.00 was reduced to 1.00 because the top two months account for 83% of scoped demand. 1.01 units of spike stock and 5.01 total branch deviation were pushed back to the DC.

- 119 : Branch location. Current min/max 1 / 2 change to 1 / 2 . Intermittent branch protection fired, so the raw active-demand mean 2.33 was reduced to 1.17 because the top two months account for 86% of scoped demand. 1.17 units of spike stock and 3.17 total branch deviation were pushed back to the DC.

- 30 : Branch location. Current min/max 3 / 4 change to 3 / 5 . The branch uses an effective demand basis of 4.80 and a variability amount of 3.60 to size the branch target, but the branch only keeps demand-only coverage. 8.00 units of deviation are intentionally not held here and are pooled to the DC instead.

- 40 : Branch location. Current min/max 3 / 4 change to 2 / 4 . The branch uses an effective demand basis of 4.17 and a variability amount of 3.32 to size the branch target, but the branch only keeps demand-only coverage. 7.00 units of deviation are intentionally not held here and are pooled to the DC instead.

iGAS USA, inc | R-404A-24LB-REFG | HP-62

- Profile: ABC A at 99% service level | season Summer | frequency 2 Weeks (14 days) | lead time 14 days | planning season Summer
- Reference window: 24 months | YOY -17.7% | rule: Year-over-year demand stayed within +/-33%, so both years are used.
- Product-group trend: Flat at 1.01x from Supplier product group
- Company total min: 110 -> 86 | company total max: 166 -> 265
- Pooled to DC: 95.00 | absorbed by DC: 95.00 | changed locations: 8 of 8

Location	Current Min	New Min	Current Max	New Max	Mean Used	Raw Mean	Total Dev	Pooled to DC	Absorbed by DC
1	27	30	59	151	32.00	32.00	24.02	0.00	95.00
115	8	6	11	12	12.00	12.00	2.86	6.00	0.00
116	20	10	25	21	20.50	20.50	6.19	13.00	0.00
117	11	9	14	19	19.08	19.08	11.69	25.00	0.00
118	22	15	28	32	31.25	31.25	12.08	25.00	0.00
119	3	2	4	3	2.20	2.20	1.98	3.00	0.00
30	10	7	13	14	14.25	14.25	5.04	10.00	0.00
40	9	7	12	13	13.00	13.00	6.20	13.00	0.00

Location reasoning:

- 1 : DC / balancing location. Current min/max 27 / 59 change to 30 / 151 . The DC min only covers location 1's own protected-cycle demand. The DC max absorbs 95.00 units of pooled branch uncertainty and is floored to policy order-up-to 83 .
- 115 : Branch location. Current min/max 8 / 11 change to 6 / 12 . The branch uses an effective demand basis of 12.00 and a variability amount of 2.86 to size the branch target, but the branch only keeps demand-only coverage. 6.00 units of deviation are intentionally not held here and are pooled to the DC instead.
- 116 : Branch location. Current min/max 20 / 25 change to 10 / 21 . The branch uses an effective demand basis of 20.50 and a variability amount of

6.19 to size the branch target, but the branch only keeps demand-only coverage. 13.00 units of deviation are intentionally not held here and are pooled to the DC instead.

- 117 : Branch location. Current min/max 11 / 14 change to 9 / 19 . The branch uses an effective demand basis of 19.08 and a variability amount of 11.69 to size the branch target, but the branch only keeps demand-only coverage. 25.00 units of deviation are intentionally not held here and are pooled to the DC instead.

- 118 : Branch location. Current min/max 22 / 28 change to 15 / 32 . The branch uses an effective demand basis of 31.25 and a variability amount of 12.08 to size the branch target, but the branch only keeps demand-only coverage. 25.00 units of deviation are intentionally not held here and are pooled to the DC instead.

- 119 : Branch location. Current min/max 3 / 4 change to 2 / 3 . The branch uses an effective demand basis of 2.20 and a variability amount of 1.98 to size the branch target, but the branch only keeps demand-only coverage. 3.00 units of deviation are intentionally not held here and are pooled to the DC instead.

- 30 : Branch location. Current min/max 10 / 13 change to 7 / 14 . The branch uses an effective demand basis of 14.25 and a variability amount of 5.04 to size the branch target, but the branch only keeps demand-only coverage. 10.00 units of deviation are intentionally not held here and are pooled to the DC instead.

- 40 : Branch location. Current min/max 9 / 12 change to 7 / 13 . The branch uses an effective demand basis of 13.00 and a variability amount of 6.20 to size the branch target, but the branch only keeps demand-only coverage. 13.00 units of deviation are intentionally not held here and are pooled to the DC instead.

iGAS USA, inc | R-421A-25LB-REFG | STD CYLINDER DYNATEMP CHOICE

- Profile: ABC B at 92% service level | season Summer | frequency 2 Weeks (14 days) | lead time 14 days | planning season Summer
- Reference window: 24 months | YOY -28.1% | rule: Year-over-year demand stayed within +/-33%, so both years are used.
- Product-group trend: Flat at 1.01x from Supplier product group
- Company total min: 8 -> 17 | company total max: 21 -> 41
- Pooled to DC: 15.19 | absorbed by DC: 15.19 | changed locations: 7 of 8

Location	Current Min	New Min	Current Max	New Max	Mean Used	Raw Mean	Total Dev	Pooled to DC	Absorbed by DC
1	2	4	10	21	3.60	3.60	1.91	0.00	15.19
115	2	3	3	5	4.92	4.92	2.21	3.00	0.00
116	0	2	1	3	2.29	2.29	1.69	1.00	0.00
117	1	3	2	5	4.60	4.60	2.85	4.00	0.00
118	0	1	0	1	0.08	1.00	0.08	0.93	0.00
119	0	0	0	0	0.00	0.00	0.00	0.00	0.00
30	1	3	2	5	5.33	5.33	3.10	4.00	0.00
40	2	1	3	1	0.25	1.50	0.96	2.26	0.00

Location reasoning:

- 1 : DC / balancing location. Current min/max 2 / 10 change to 4 / 21 . The DC min only covers location 1's own protected-cycle demand. The DC max absorbs 15.19 units of pooled branch uncertainty and is floored to policy order-up-to 7 .
- 115 : Branch location. Current min/max 2 / 3 change to 3 / 5 . The branch uses an effective demand basis of 4.92 and a variability amount of 2.21 to size the branch target, but the branch only keeps demand-only coverage. 3.00 units of deviation are intentionally not held here and are pooled to the DC instead.
- 116 : Branch location. Current min/max 0 / 1 change to 2 / 3 . The branch uses an effective demand basis of 2.29 and a variability amount of 1.69

to size the branch target, but the branch only keeps demand-only coverage. 1.00 units of deviation are intentionally not held here and are pooled to the DC instead.

- 117 : Branch location. Current min/max 1 / 2 change to 3 / 5 . The branch uses an effective demand basis of 4.60 and a variability amount of 2.85 to size the branch target, but the branch only keeps demand-only coverage. 4.00 units of deviation are intentionally not held here and are pooled to the DC instead.

- 118 : Branch location. Current min/max 0 / 0 change to 1 / 1 . Intermittent branch protection fired, so the raw active-demand mean 1.00 was reduced to 0.08 because it only sold in 1 of 12 scoped months, ADI is 12.0, the top two months account for 100% of scoped demand. 0.93 units of spike stock and 0.93 total branch deviation were pushed back to the DC.

- 119 : Branch location. Current min/max 0 / 0 change to 0 / 0 . The branch uses an effective demand basis of 0.00 and a variability amount of 0.00 to size the branch target, but the branch only keeps demand-only coverage. 0.00 units of deviation are intentionally not held here and are pooled to the DC instead.

- 30 : Branch location. Current min/max 1 / 2 change to 3 / 5 . The branch uses an effective demand basis of 5.33 and a variability amount of 3.10 to size the branch target, but the branch only keeps demand-only coverage. 4.00 units of deviation are intentionally not held here and are pooled to the DC instead.

- 40 : Branch location. Current min/max 2 / 3 change to 1 / 1 . Intermittent branch protection fired, so the raw active-demand mean 1.50 was reduced to 0.25 because it only sold in 2 of 12 scoped months, ADI is 6.0, the top two months account for 100% of scoped demand. 1.26 units of spike stock and 2.26 total branch deviation were pushed back to the DC.

iGAS USA, inc | R-438A-25LB-REFG | MO99 R-22 REPLACEMENT A.C

- Profile: ABC A at 99% service level | season Summer | frequency 2 Weeks (14 days) | lead time 14 days | planning season Summer
- Reference window: 24 months | YOY -3.4% | rule: Year-over-year demand stayed within +/-33%, so both years are used.
- Product-group trend: Flat at 1.01x from Supplier product group
- Company total min: 39 -> 82 | company total max: 88 -> 303
- Pooled to DC: 132.00 | absorbed by DC: 132.00 | changed locations: 8 of 8

Location	Current Min	New Min	Current Max	New Max	Mean Used	Raw Mean	Total Dev	Pooled to DC	Absorbed by DC
1	7	18	44	168	18.75	18.75	17.08	0.00	132.00
115	3	8	4	17	16.33	16.33	8.59	19.00	0.00
116	6	14	8	30	29.75	29.75	9.96	20.00	0.00
117	13	14	17	29	29.00	29.00	11.93	24.00	0.00
118	2	4	3	7	6.58	6.58	2.89	7.00	0.00
119	0	2	1	4	3.30	3.30	3.17	6.00	0.00
30	4	10	6	22	21.42	21.42	8.77	19.00	0.00
40	4	12	5	26	25.75	25.75	17.13	37.00	0.00

Location reasoning:

- 1 : DC / balancing location. Current min/max 7 / 44 change to 18 / 168 . The DC min only covers location 1's own protected-cycle demand. The DC max absorbs 132.00 units of pooled branch uncertainty and is floored to policy order-up-to 56 .
- 115 : Branch location. Current min/max 3 / 4 change to 8 / 17 . The branch uses an effective demand basis of 16.33 and a variability amount of 8.59 to size the branch target, but the branch only keeps demand-only coverage. 19.00 units of deviation are intentionally not held here and are pooled to the DC instead.
- 116 : Branch location. Current min/max 6 / 8 change to 14 / 30 . The branch uses an effective demand basis of 29.75 and a variability amount of

9.96 to size the branch target, but the branch only keeps demand-only coverage. 20.00 units of deviation are intentionally not held here and are pooled to the DC instead.

- 117 : Branch location. Current min/max 13 / 17 change to 14 / 29 . The branch uses an effective demand basis of 29.00 and a variability amount of 11.93 to size the branch target, but the branch only keeps demand-only coverage. 24.00 units of deviation are intentionally not held here and are pooled to the DC instead.

- 118 : Branch location. Current min/max 2 / 3 change to 4 / 7 . The branch uses an effective demand basis of 6.58 and a variability amount of 2.89 to size the branch target, but the branch only keeps demand-only coverage. 7.00 units of deviation are intentionally not held here and are pooled to the DC instead.

- 119 : Branch location. Current min/max 0 / 1 change to 2 / 4 . The branch uses an effective demand basis of 3.30 and a variability amount of 3.17 to size the branch target, but the branch only keeps demand-only coverage. 6.00 units of deviation are intentionally not held here and are pooled to the DC instead.

- 30 : Branch location. Current min/max 4 / 6 change to 10 / 22 . The branch uses an effective demand basis of 21.42 and a variability amount of 8.77 to size the branch target, but the branch only keeps demand-only coverage. 19.00 units of deviation are intentionally not held here and are pooled to the DC instead.

- 40 : Branch location. Current min/max 4 / 5 change to 12 / 26 . The branch uses an effective demand basis of 25.75 and a variability amount of 17.13 to size the branch target, but the branch only keeps demand-only coverage. 37.00 units of deviation are intentionally not held here and are pooled to the DC instead.

iGAS USA, inc | R-507-25LB-REFG | AZ-50 R-507A

- Profile: ABC B at 92% service level | season Summer | frequency 2 Weeks (14 days) | lead time 14 days | planning season Summer
- Reference window: 12 months | YOY -74.4% | rule: Year-over-year demand moved outside +/-33%, so the model uses the last 12 months.
- Product-group trend: Flat at 1.01x from Supplier product group
- Company total min: 8 -> 7 | company total max: 29 -> 24
- Pooled to DC: 15.10 | absorbed by DC: 15.10 | changed locations: 4 of 8

Location	Current Min	New Min	Current Max	New Max	Mean Used	Raw Mean	Total Dev	Pooled to DC	Absorbed by DC
1	4	4	21	18	4.00	4.00	0.03	0.00	15.10
115	0	0	0	0	0.00	0.00	0.00	0.00	0.00
116	0	0	1	1	0.00	0.00	0.00	0.00	0.00
117	1	1	2	1	0.33	2.00	0.34	1.68	0.00
118	2	0	3	1	0.00	0.00	0.00	0.00	0.00
119	1	2	2	3	2.67	16.00	-0.32	13.42	0.00
30	0	0	0	0	0.00	0.00	0.00	0.00	0.00
40	0	0	0	0	0.00	0.00	0.00	0.00	0.00

Location reasoning:

- 1 : DC / balancing location. Current min/max 4 / 21 change to 4 / 18 . The DC min only covers location 1's own protected-cycle demand. The DC max absorbs 15.10 units of pooled branch uncertainty and is floored to policy order-up-to 4 .
- 115 : Branch location. Current min/max 0 / 0 change to 0 / 0 . The branch uses an effective demand basis of 0.00 and a variability amount of 0.00 to size the branch target, but the branch only keeps demand-only coverage. 0.00 units of deviation are intentionally not held here and are pooled to the DC instead.
- 116 : Branch location. Current min/max 0 / 1 change to 0 / 1 . The branch uses an effective demand basis of 0.00 and a variability amount of 0.00

to size the branch target, but the branch only keeps demand-only coverage. 0.00 units of deviation are intentionally not held here and are pooled to the DC instead.

- 117 : Branch location. Current min/max 1 / 2 change to 1 / 1 . Intermittent branch protection fired, so the raw active-demand mean 2.00 was reduced to 0.33 because it only sold in 1 of 6 scoped months, ADI is 6.0, the top two months account for 100% of scoped demand. 1.68 units of spike stock and 1.68 total branch deviation were pushed back to the DC.
- 118 : Branch location. Current min/max 2 / 3 change to 0 / 1 . The branch uses an effective demand basis of 0.00 and a variability amount of 0.00 to size the branch target, but the branch only keeps demand-only coverage. 0.00 units of deviation are intentionally not held here and are pooled to the DC instead.
- 119 : Branch location. Current min/max 1 / 2 change to 2 / 3 . Intermittent branch protection fired, so the raw active-demand mean 16.00 was reduced to 2.67 because it only sold in 1 of 6 scoped months, ADI is 6.0, the top two months account for 100% of scoped demand. 13.42 units of spike stock and 13.42 total branch deviation were pushed back to the DC.
- 30 : Branch location. Current min/max 0 / 0 change to 0 / 0 . The branch uses an effective demand basis of 0.00 and a variability amount of 0.00 to size the branch target, but the branch only keeps demand-only coverage. 0.00 units of deviation are intentionally not held here and are pooled to the DC instead.
- 40 : Branch location. Current min/max 0 / 0 change to 0 / 0 . The branch uses an effective demand basis of 0.00 and a variability amount of 0.00 to size the branch target, but the branch only keeps demand-only coverage. 0.00 units of deviation are intentionally not held here and are pooled to the DC instead.